

Deepak Mallampati

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Kent, OH, USA - open to Paris relocation; F-1 OPT, France Talent Passport pathway

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) building agentic LLM workflows, RAG systems, and predictive ML pipelines end-to-end - from data preprocessing through agent design, evaluation, and FastAPI/Docker deployment. Track record: ~35% retrieval precision gain, >30% hallucination reduction, 15-20% recall lift on high-risk categories, ~30% post-deploy defect reduction. Comfortable owning customer-facing implementation: translating business problems into structured AI agents, schema-validated multi-agent pipelines, and shipping the result to production.

SKILLS AND CORE COMPETENCIES

Agentic AI Workflows • RAG + Grounding • Multi-Agent Systems • LLM Application Development • Python + SQL •

FastAPI / Docker • Customer-Facing Delivery • Structured Outputs + Evaluation • System Integration • Prototyping

Languages and Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++, PHP

LLM and GenAI: Agentic workflows, RAG, structured outputs, grounding, evaluation pipelines, prompt engineering, embeddings, vector search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Infra and DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, structured logging, observability

Domains: Applied AI, Enterprise AI, Healthcare AI (HIPAA-conscious), Workflow Automation, Customer-Facing AI Deployment

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - USA, Healthcare Technology / AI Consulting

- Designed **agentic LLM workflows** for multi-step healthcare queries (eligibility, care navigation, documentation) with structured reasoning, tool discipline, and grounding; **+~25% agent response stability**.
- Built **RAG systems** for clinical knowledge retrieval - recursive semantic chunking + transformer embeddings; **+~35% retrieval precision, >30% irrelevant-retrieval reduction, >90% grounded alignment**.
- Owned end-to-end implementation: **FastAPI/Flask** services containerized with **Docker**, structured logging, load simulation; **~30% fewer post-deployment defects**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for no-show + care-engagement scoring; **+15-20% recall on high-risk categories at >90% precision**.
- Maintained HIPAA-conscious governance: de-identification, lineage docs, evaluation audit trails, system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database and DevOps Performance Engineer (Intern) - Hyderabad, India, Oil and Gas ERP

- Built **Jenkins CI/CD pipelines** for schema updates and stored-procedure deployments; **>30% deploy-error reduction, ~35-40% faster release cycle** via structured validation and rollback checkpoints.
- Tuned **T-SQL stored procedures** and Order-to-Cash ERP business logic (contracts, nominations, allocations, invoicing); **~20% query-time and ~25% data-retrieval latency reduction**.
- Partnered with financial / ERP business owners on reconciliation queries, RBAC, and audit logging in a compliance-sensitive environment.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Hyderabad, India, Nonprofit Tech

- Built a **transformer-based video summarization and highlight system** over 5,000+ sessions; **60-70% manual-review reduction, ~85% alignment** with human-curated highlights.
- Implemented document Q&A with **RAG**; **~30% hallucination reduction, >85% contextual answer accuracy**; Flask API for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing and Fraud Risk Intelligence System

Multi-agent pipeline (Intake to Validation to Consistency to Duplicate Detection to Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation; ANN similarity search for duplicate claims; explainable risk scoring with audit-ready traces.

Python, multi-agent orchestration, RAG, vector search, JSON schema validation

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped, Chrome Web Store

Full-stack browser extension built solo - Manifest V3, content scripts, service workers, `captureVisibleTab` for panel capture. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama). Multi-section capture for tall panels with coordinate remapping; 7-day cache; per-domain selectors for 29 sites; privacy policy + narrowed host permissions.

TypeScript, Chrome MV3, multi-provider AI APIs

Dream Decoder - Multimodal Creative Intelligence Platform

FastAPI + React app that interprets dreams and synthesizes dreamscapes via coordinated multimodal APIs. **Intermediate structured prompt transformation layers** - ~30% contextual alignment lift and ~25-30% first-pass image success rate lift.

FastAPI, React, multimodal AI, structured prompt orchestration

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025